

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. INTRODUCTION TO PERFORMANCE TESTING

Performance testing is a critical phase in ensuring that ComicCraft operates reliably, responds within acceptable timeframes, and gracefully handles network constraints or external API failures. Unlike traditional web applications that only query an internal database, ComicCraft depends heavily on external Generative AI services (Google Gemini and Hugging Face Stable Diffusion). Therefore, testing focuses heavily on response latency, concurrency handling, and automated fallback resilience.

3. TESTING OBJECTIVES

- To measure the average execution time of the end-to-end comic generation pipeline.
- To evaluate the effectiveness and speed of the Dual-Engine AI image fallback mechanism.
- To test the stability of the FastAPI backend under local asynchronous routing.
- To verify the efficiency of the FPDF compilation module in generating multi-page PDF files without memory leaks or formatting delays.

4. TEST SCENARIOS AND METRICS

Test Scenario 1: Standard End-to-End Generation Workflow

- **Input Parameters:** Standard prompt ("A funny day in the park"), Character Name ("Max"), Setting ("Park"), Tone ("Funny"), Art Style ("Comic Book").
- **Expected Result:** Successful generation of structural outline, full narrative, and panel images within 45 seconds.

- **Actual Result:** Completed in approximately **32.4 seconds**. The Google Gemini Flash model responded in ~3s, Gemini Pro in ~4s, and the Pollinations/Hugging Face image generation took ~22s cumulatively.

Test Scenario 2: Dual-Engine API Failover Latency

- **Condition:** Triggered intentionally when Engine 1 (Pollinations AI) returns an HTTP 402 (Payment Required) or HTTP 503 (Service Unavailable) error.
- **Expected Result:** The system must instantly catch the exception and switch to Engine 2 (Hugging Face Stable Diffusion) without throwing a 500 Internal Server Error to the user.
- **Actual Result:** Failover execution overhead added only **4.5 seconds** to the total request lifecycle. The application successfully recovered and fetched the image via the backup route.

Test Scenario 3: Extreme Network / DNS Block Resilience

- **Condition:** Simulated local ISP DNS block ([Errno 11001] getaddrinfo failed) where all external image APIs are unreachable.
- **Expected Result:** The system must trigger the ultimate fallback (Python PIL Comic Caption Panel) to prevent application crashes and ensure PDF export continuity.
- **Actual Result:** The fallback image was rendered locally in **0.12 seconds**. The user experienced zero downtime, and the PDF compiled successfully with the professional yellow caption box.

Test Scenario 4: PDF Export Compilation Speed

- **Input:** Compiling text data, metadata, and 2 to 5 high-resolution JPEG panel images into a downloadable document.
- **Expected Result:** PDF generation must occur instantly upon user request.
- **Actual Result:** FPDF successfully compiled and saved the multi-page layout in **0.85 seconds**, storing the output securely in the `static/exports/` directory.

5. CONCURRENCY AND LOAD OBSERVATIONS

- **Uvicorn ASGI Performance:** Tested using local rapid sequential requests. FastAPI managed concurrent form submissions efficiently using asynchronous routing. While external API rate limits (from free-tier AI providers) restricted heavy parallel processing, the local application environment remained stable and responsive.

6. CONCLUSION OF TESTING

The performance testing phase validated that ComicCraft is robust, fault-tolerant, and production-ready for evaluation. The implementation of multi-layered exception handling ensures that external network instabilities never compromise the user experience or prevent successful PDF generation.